Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Four persons are selected at random from a group of 3 men, 2 women and 4 children. The chance that exactly 2 of them are children, is

(A) $\frac{9}{21}$

(B) $\frac{10}{23}$

(C) $\frac{11}{24}$

(D) $\frac{10}{21}$

Q.2

One Indian and four American men and their wives are to be seated randomly around a circular table. Then the conditional probability that the Indian man is seated adjacent to his wife given that each American man is seated adjacent to his wife is

(A) $\frac{1}{2}$

(B) $\frac{1}{3}$

(C) $\frac{2}{5}$

(D) $\frac{1}{5}$

Q.3

A and B toss a die alternatively till one of them gets a six and wins the game. If A begins the game, then the probability of B winning the game is

(A) $\frac{5}{11}$

(B) $\frac{6}{11}$

(C) $\frac{7}{11}$

(D) $\frac{4}{11}$

Q.4

A is a 3×3 matrix with entries from the set $\{-1, 0, 1\}$. Then the probability that A is neither symmetric nor skew-symmetric is

(A) $\frac{3^9-3^6-3^3+1}{3^9}$

(B) $\frac{3^9+3^6-3^3+1}{3^9}$

(C) $\frac{3^9-3^6+3^3+1}{3^9}$

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(D) $\frac{1}{2}$

Q.5

If E and F are events with $P(E) \leq P(F)$ and $P(E \cap F) > 0$, then

(A) occurrence of $E \Rightarrow$ occurrence of F

(B) occurrence of $F \Rightarrow$ occurrence of E

(C) non-occurrence of $E \Rightarrow$ non-occurrence of F

(D) none of these

Q.6

There are n persons sitting in a row. Two of them are selected at random. The probability that two selected persons are not together is

(A) $\frac{2}{n}$

(B) $1 - \frac{2}{n}$

(C) $\frac{n(n-1)}{(n+1)(n+2)}$

(D) none of these

Q.7

A box contains 24 identical balls of which 12 are white and 12 are black, let the balls are drawn at random from the box, one at a time with replacement, then the probability that a white ball is drawn for the 4th time on the 7th draw is

(A) $\frac{5}{64}$

(B) $\frac{27}{32}$

(C) $\frac{5}{32}$

(D) $\frac{1}{2}$

Q.8

The probabilities of three events A, B and C are $P(A) = 0.6, P(B) = 0.4, P(C) = 0.5$ also $P(A \cup B) = 0.8, \quad P(A \cap C) = 0.3, P(A \cup B \cup C) \geq 0.85, P(A \cap B \cap C) = 0.2$ and $P(B \cap C) = p_1$. Then

(A) $p_1 \leq 0.1$

(B) $p_1 \geq 0.4$

(C) $0.2 \leq p_1 \leq 0.35$

(D) none of these

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Q.9

If A and B are two mutually exclusive events, then

(A) $P(A) \leq P(\bar{B})$

(B) $P(A) > P(\bar{B})$

(C) $P(\bar{A}) < P(\bar{B})$

(D) none of these

Q.10

One hundred identical coins, each with probability p of showing heads are tossed once. If $0 < p < 1$ and the probability of heads showing on 50 coins is equal to that of heads showing on 51 coins, the value of p is

(A) $\frac{1}{2}$

(B) $\frac{51}{101}$

(C) $\frac{49}{101}$

(D) none of these

Q.11

A bag contains three coins, one of which has head on both sides, another is a biased coin that shows up heads 90% of the time and the third one is an unbiased coin. A coin is taken out from the bag at random and tossed. If it shows up a head, then the probability that it is the unbiased coin, is:

(A) $\frac{3}{8}$

(B) $\frac{5}{12}$

(C) $\frac{5}{24}$

(D) $\frac{1}{3}$

Q.12

A fair coin is tossed repeatedly. If tail appears on first four tosses, then the probability of head appearing on fifth toss equals

(A) $1/2$

(B) $1/32$

(C) $31/32$

(D) $1/5$

Q.13

An event succeeds twice as many times as it fails, then the probability that in the next 5 trials it will succeed at least 3 times, is

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(A) $\frac{64}{81}$

(B) $\frac{65}{81}$

(C) $\frac{66}{81}$

(D) $\frac{67}{81}$

Q.14

If A and B are two independent events such that $P\left(A'\right)=\frac{7}{10}, P\left(B'\right)=\alpha$ and $P(A \cup B)=\frac{8}{10}$, then α is

(A) $\frac{2}{7}$

(B) $\frac{5}{7}$

(C) 1

(D) none of these

Q.15

The mean and variance of a binomial variable X are 2 and 1 respectively. The probability that X takes values greater than 1, is

(A) $\frac{5}{16}$

(B) $\frac{9}{16}$

(C) $\frac{11}{16}$

(D) none of these

Q.16

Let A and B be two events such that $P(\overline{A \cup B})=\frac{1}{6}, P(A \cap B)=\frac{1}{4}$ and $P(\bar{A})=\frac{1}{4}$ and, where $\bar{A}$ stands for complement of event A. Then events A and B are

(A) independent but not equally likely

(B) mutually exclusive and independent

(C) equally likely and mutually exclusive

(D) equally likely but not independent

Q.17

It has been found that if A and B play a game 12 times, A wins 6 times, B wins 4 times and they draw twice. A and B take part in a series of 3 games. The probability that they will win alternately is

(A) $\frac{5}{72}$

(B) $\frac{5}{36}$

(C) $\frac{19}{27}$

(D) none of these

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Q.18

If two events A and B such that $P\left(A'\right)=0.3, P(B)=0.5$ and $P(A \cap B)=0.3$, then $P\left(B \mid A \cup B'\right)$ is

(A) 0.6

(B) 0.32

(C) 0.31

(D) 0.28

Q.19

A bag contains some white and some black balls, all combinations of balls being equally likely. The total number of balls in the bag is 10 . If three balls are drawn at random without replacement and all of them are found to be black, the probability that the bag contains 1 white and 9 black balls is

(A) $\frac{14}{55}$

(B) $\frac{12}{55}$

(C) $\frac{2}{11}$

(D) $\frac{8}{55}$

Q.20

The probability that a student is not a swimmer is $\frac{1}{5}$. The probability that out of 5 students exactly 4 are swimmer is

(A) $\left(\frac{4}{5}\right)^3$

(B) $\left(\frac{4}{5}\right)^4$

(C) $\left(\frac{1}{5}\right)$

(D) none of these

Q.21

The sum of all possible values of $P(A)$ if event A is independent of itself is

Q.22

A random variable X has the probability distribution:
\begin{array} {|c|c|c|c|c|c|c|c|} \ \hline x: & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ \ \hline p(x): & 0.15 & 0.23 & 0.12 & 0.10 & 0.20 & 0.08 & 0.07 & 0.05 \\ \ \hline \ \end{array}
For the events $E=\{X$ is prime number $\}$ and $F=\{X<4\}$, the probability $P(E \cup F)$ is p/100 then p=

Q.23

A signal which can be green or red with probability $\frac{4}{5}$ and $\frac{1}{5}$ respectively, is received by station A and then transmitted to station B. The probability of each station receiving the signal correctly is $\frac{3}{4}$. If the signal received at

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station B is green, then the probability that the original signal was green is p/q where p and q are positive coprime integers, then p+q is

Q.24

Two unbiased die are rolled together.
Let $A = \{(a, b) : a + b = 11\}$ and $B = \{(a, b) : a \neq 5\}$, then $P(A \mid B)$ equals (mark reciprocal of answer)

Q.25

If the probability that the product of the outcomes of three rolls of a fair dice is a prime number is p, then the value of $1/(4p)$ is.

Q.26

A drawer contains a mixture of red socks and blue socks, at most 17 in all. It so happens that when two socks are selected randomly without replacement, there is a probability of exactly $1/2$ that both are red or both are blue. The largest possible number of red socks in the drawer that is consistent with this data is.

Q.27

A bag contains $n + 1$ coins. It is known that one of these coins shows heads on both sides, whereas the other coins are fair. One coin is selected at random and tossed. If the probability that toss results in heads is $7/12$, then find the value of n.

Q.28

If the standard deviation of the binomial distribution $(q + p)^{16}$ is 2, then mean of the distribution is

Q.29

The probability distribution of a random variable X is given as follows:
\begin{array}{|c|c|c|c|c|c|} \hline x & 1 & 2 & 3 & 2 k & 3 k & 5 k \\ \hline P(x=x) & 1 / 2 & 1 / 5 & 3 / 25 & 1 / 10 & 1 / 25 & 1 / 25 \\ \hline \end{array}
If $E(X) = 2.3$, then value of k is

Q.30

Thirty-two players ranked 1 to 32 are playing in a knockout tournament. Assume that in every match between any two players the better ranked player wins, the probability that ranked 1 and ranked 2 players are winner and runner up respectively is p, then the value of $[2/p]$ is, where [.] represents the greatest integer function

Answers & Solutions

Q.1 Answer:

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$\frac{10}{21}$

Solution:

Two children can be chosen in $^4C_2$ **ways**

and 2 other person from 5 men and women can be chosen in $^5C_2$ **ways**

$\Rightarrow \quad$ Required probability $= \frac{^4C_2 \times ^5C_2}{^9C_4} = \frac{10}{21}$

Q.2 Answer:

$\frac{2}{5}$

Solution:

Let E $=$ **event when each American man is seated adjacent to his wife. A = event when Indian man is seated adjacent to his wife.**

So, there are 5 couples out of which 4 are American and 1 is Indian.

Now, $n(A \cap E) = 4!(2!)^5$**. (Each couple is taken together as one entity, so we have total 5 things to be arranged in a circle, which can be done in** $(5-1)! = 4!$ **number of ways. Also, each couple as one entity can be seated in 2 ways.)**

And $n(E) = 5!(2!)^4$ **(Each American couple is taken as one entity, and an Indian man and woman separately. Hence, the arrangement of 6 things in a circle** $= 5!$**. Also each American couple as one entity can be seated in 2 ways.)**

So, $P(A \mid E) = \frac{n(A \cap E)}{n(E)}$

$\Rightarrow \mathrm{P(A \mid E)} = \frac{4!(2!)^5}{5!(2!)^4}$

$\Rightarrow \mathrm{P(A \mid E)} = \frac{2}{5}$

Q.3 Answer:

$\frac{5}{11}$

Solution:

Let S denote the success (getting a '6') and F **denote the failure (not getting a '6').**

Thus, $\mathrm{P(S)} = \frac{1}{6} = \mathrm{p}, \mathrm{P(F)} = \frac{5}{6} = \mathrm{q}$

$\mathrm{P}($**A wins in first throw** $) = \mathrm{P(S)} = \mathrm{p}$

$\mathrm{P}($**A wins in third throw** $) = \mathrm{P(FFS)} = \mathrm{qqp}$

$\mathrm{P}($**A wins in fifth throw** $) = \mathrm{P(FFFFS)} = \mathrm{qqqqp}$

So, $\mathrm{P}($**A wins** $) = \mathrm{p} + \mathrm{qqp} + \mathrm{qqqqp} + \ldots$

$= \mathrm{p}\left(1 + \mathrm{q}^2 + \mathrm{q}^4 + \ldots\right)$

$= \frac{\mathrm{p}}{1-\mathrm{q}^2} = \frac{\frac{1}{6}}{1-\frac{25}{36}} = \frac{6}{11}$

$\mathrm{P}($**B wins**$) = 1 - \mathrm{P}($**A wins**$)$

$\mathrm{P}($**B wins**$) = 1 - \frac{6}{11} = \frac{5}{11}$

So, $\mathrm{P}($**A wins**$) = \frac{6}{11}, \mathrm{P}($**B wins**$) = \frac{5}{11}$

Q.4 Answer:

$\frac{3^9 - 3^6 - 3^3 + 1}{3^9}$

Solution:

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Total number of matrices $= 3^9$

Number of symmetric matrices $= 3^6$

Number of skew symmetric matrices $= 3^3$

The zero matrix is both symmetric and skew-symmetric the required probability is

$= \frac{3^9-3^6-3^3+1}{3^9}$

Q.5 Answer:

none of these

Solution:

Consider the following events in a single throw of a die.

$E =$ **getting a multiple of 3**

$F =$ **getting an even number.**

We have, $P(E) = 1/3$ **and** $P(F) = 1/2$

Clearly, $P(E) \leq P(F)$ **and** $P(E \cap F) = 1/6 > 0$

But none of the options (a), (b) and (c) is true.

Q.6 Answer:

$1 - \frac{2}{n}$

Solution:

The total number of ways of selecting two persons out of n **is** ${}^nC_2 = \frac{n(n-1)}{2}$**. The number of ways in which two selected persons are together is** $(n-1)$**.**

So, the probability that the selected persons are together is $\frac{n-1}{\frac{n(n-1)}{2}} = \frac{2}{n}$

Hence, required probability $= 1 - \frac{2}{n}$

Q.7 Answer:

$\frac{5}{32}$

Solution:

In any trial, $p($ **getting white ball)** $= \frac{1}{2}$**;** p **(getting black ball)** $= \frac{1}{2}$

Required probability $= P(WWWBBB$ **any permutation)** W

$= 20 \times \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^3 \times \frac{1}{2} = \frac{5}{32}$

Q.8 Answer:

$0.2 \leq p_1 \leq 0.35$

Solution:

$P(A \cap B) = P(A) + P(B) - P(A \cup B) = 0.6 + 0.4 - 0.8 = .2$

Now, $0.85 \leq P(A \cup B \cup C) \leq 1$

$\Rightarrow 0.85 \leq P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C) \leq 1$

$\Rightarrow 0.2 \leq p_1 \leq 0.35$

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Q.9 Answer:

$P(A) \leq P(\bar{B})$

Solution:

Since A and B are mutually exclusive events, therefore

$$A \cap B = \varphi \Rightarrow A \subseteq \bar{B} \quad \text{and } B \subseteq \bar{A}$$

$$\Rightarrow \quad P(A) \leq P(\bar{B}) \text{ and } P(B) \leq P(\bar{A})$$

Q.10 Answer:

$\frac{51}{101}$

Solution:

Let X be the number of coins showing heads.

Then X follows a binomial distribution with parameters $n = 100$ and p.

Since $P(X = 50) = P(X = 51)$, we get, ${}^{100}C_{50}p^{50}(1-p)^{50} = {}^{100}C_{51}p^{51}(1-p)^{49}$

$\Rightarrow \frac{100!}{50!50!} \cdot \frac{51!49!}{100!} = \frac{p}{1-p} \Rightarrow \quad \frac{51}{50} = \frac{p}{1-p} \Rightarrow 51 - 51p = 50p \Rightarrow \quad p = \frac{51}{101}$

Q.11 Answer:

$\frac{5}{24}$

Solution:

Let $\mathrm{E}_1, \mathrm{E}_2, \mathrm{E}_3$ be the following events.

E_1 : coin with two heads is chosen

E_2 : coin with 90% chance of getting head is chosen E_3 : unbiased coin is chosen

And let A be the event that a head occurs when the coin is tossed.

We have $\mathrm{P}\left(\mathrm{E}_1\right) = \mathrm{P}\left(\mathrm{E}_2\right) = \mathrm{P}\left(\mathrm{E}_3\right) = \frac{1}{3}\ \mathrm{P}\left(\mathrm{AlE}_1\right) = 1, \mathrm{P}\left(\mathrm{AlE}_2\right) = 0.9, \mathrm{P}\left(\mathrm{AlE}_3\right)^3 = \frac{1}{2}$

By the Bayes rule

$$\mathrm{P}\left(\mathrm{E}_3 \mid \mathrm{A}\right)$$

$$= \frac{P(E_3)P(A|E_3)}{P(E_1)P(A|E_1)+P(E_2)P(A|E_2)+P(E_3)P(A|E_3)}$$

$$= \frac{\frac{1}{2}}{1+\frac{9}{10}+\frac{1}{2}}$$

$$= \frac{5}{10+9+5} = \frac{5}{24}$$

Q.12 Answer:

$1/2$

Solution:

when the coin is flipped fifth time, there is a chance of getting head or tail. Therefore the chance of getting head $=$ chance of getting tail so probability of getting head on fifth toss is $1/2$. it doesn't depend on first four outcomes.

Q.13 Answer:

$\frac{64}{81}$

Solution:

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$p = 2q$ ...(i)

$p + q = 1$ ....(ii)

Solving (i) and (ii), $p = \frac{2}{3}, q = \frac{1}{3}$

$\therefore$ Required probability $= {}^5C_3 \left(\frac{2}{3}\right)^3 \left(\frac{1}{3}\right)^2 + {}^5C_4 \left(\frac{2}{3}\right)^4 \left(\frac{1}{3}\right) + {}^5C_5 \left(\frac{2}{3}\right)^5 \left(\frac{1}{3}\right)^0 = \frac{64}{81}$

Q.14 Answer:

$\frac{2}{7}$

Solution:

Since A, B are independent

$\therefore \quad P(A' \cap B') = P(A')\,P(B')$

$\Rightarrow \quad 1 - P(A \cup B) = P(A')\,P(B') \Rightarrow \quad 1 - \frac{8}{10} = \left(\frac{7}{10}\right)\alpha \quad \alpha = \frac{2}{7}$

Q.15 Answer:

$\frac{11}{16}$

Solution:

Given mean $np = 2$ ...(i)

and variance $npq = 1$ ...(ii)

Dividing (ii) by (i), then $q = \frac{1}{2}$

$\therefore p = 1 - q = \frac{1}{2}$

From (i), $n \times \frac{1}{2} = 2, \quad \therefore n = 4$

The binomial distribution is $\left(\frac{1}{2} + \frac{1}{2}\right)^4$

Now, $P(X > 1) = P(X = 2) + P(X = 3) + P(X = 4)$

$= {}^4C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^2 + {}^4C_3 \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^3 + {}^4C_4 \left(\frac{1}{2}\right)^4 = \frac{6+4+1}{16} = \frac{11}{16}$

Q.16 Answer:

independent but not equally likely

Solution:

$P(\overline{A \cup B}) = \frac{1}{6}, P(A \cap B) = \frac{1}{4} \quad P(\bar{A}) = \frac{1}{4}$

$\Rightarrow$

$$P(A) = \frac{3}{4}, P(\overline{A \cup B}) = 1 - P(A \cup B)$$

$$= 1 - P(A) - P(B) + P(A \cap B)$$

$$\Rightarrow \frac{1}{6} = \frac{1}{4} - P(B) + \frac{1}{4}$$

$$\Rightarrow P(B) = \frac{1}{3}$$

Since $P(A \cap B) = P(A)P(B)$ and $P(A) \neq P(B)$ therefore A and B are independent but not equally likely.

Q.17 Answer:

$\frac{5}{36}$

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Solution:

The probability of A winning in a game $= P(A) = \frac{6}{12} = \frac{1}{2}$
The probability of B winning in a game $= P(B) = \frac{4}{12} = \frac{1}{3}$
The required probability $= P(A \cap B \cap A) + P(B \cap A \cap B)$
$= P(A) \cdot P(B) \cdot P(A) + P(B) \cdot P(A) \cdot P(B) = \frac{1}{2} \cdot \frac{1}{3} \cdot \frac{1}{2} + \frac{1}{3} \cdot \frac{1}{2} \cdot \frac{1}{3} = \frac{5}{36}$

Q.18 Answer:

0.6

Solution:

We have, $P(A \cup B') = P(A) + P(B') - P(A \cap B')$
$= [1 - P(A')] + [1 - P(B)] - [P(A) - P(A \cap B)]$
$= (1 - 0.3) + (1 - 0.5) - (1 - 0.3) = 0.5$
Now,

$$P(B \mid A \cup B') = \frac{P[B \cap (A \cup B')]}{P(A \cup B')} = \frac{P[(B \cap A) \cup (B \cap B')}{P(A \cup B')}$$
$$= \frac{P(A \cap B)}{P(A \cup B')} = \frac{0.3}{0.5} = 0.6$$

Q.19 Answer:

$\frac{14}{55}$

Solution:

Let E_i denote the event that the bag contains i black
and $(10 - i)$ white balls $(i = 0, 1, 2, \ldots\ldots, 10)$
Let A denote the event that the three balls drawn at random from the bag are black.
We have, $P(E_i) = \frac{1}{11} \quad (i = 0, 1, 2, \ldots\ldots, 10)$
$P(A \mid E_i) = 0$ for $i = 0, 1, 2$ and $\quad P(A \mid E_i) = \frac{{}^{i}C_3}{{}^{10}C_3}$ for $i \geq 3$
Now, $P(A) = \sum_{i=0}^{10} P(E_i) P(A \mid E_i) = \frac{1}{11} \times \frac{1}{{}^{10}C_3} \left[{}^{3}C_3 + {}^{4}C_3 + \ldots\ldots + {}^{10}C_3\right]$
But ${}^{3}C_3 + {}^{4}C_3 + {}^{5}C_3 + \ldots\ldots + {}^{10}C_3$
$= {}^{4}C_4 + {}^{4}C_3 + {}^{5}C_3 + \ldots\ldots + {}^{10}C_3 \left[{}^{3}C_3 = 1 = {}^{4}C_4\right]$
$= {}^{5}C_4 + {}^{5}C_3 + {}^{6}C_3 + \ldots\ldots + {}^{10}C_3$
$= {}^{6}C_4 + {}^{6}C_3 + \ldots\ldots + {}^{10}C_3 = \ldots\ldots = {}^{11}C_4$
Thus, $P(A) = \frac{{}^{11}C_4}{11 \times {}^{10}C_3} = \frac{1}{4}$
$P(E_9 \mid A) = \frac{P(E_9)P(A|E_9)}{P(A)} = \frac{\frac{1}{11} \frac{{}^{9}C_3}{{}^{10}} C_3}{1/4} = \frac{14}{55}$

Q.20 Answer:

$\left(\frac{4}{5}\right)^4$

Solution:

The repeated selection of students who are swimmers are Bernoulli trials. Let X denote the number of students, out of 5 students, who are swimmers.

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Probability of students who are not swimmers, $q = \frac{1}{5}$

$\therefore \mathrm{p} = 1 - \mathrm{q} = 1 - \frac{1}{5} = \frac{4}{5}$

Clearly, X **has a binomial distribution with** $\mathrm{n} = 5$ **and** $\mathrm{p} = \frac{4}{5}$

$\mathrm{P}(\mathrm{X} = \mathrm{x}) = {}^{n}\mathrm{C}_{x} q^{\mathrm{n}-\mathrm{x}} \mathrm{p}^{x} = {}^{5}\mathrm{C}_{x} \left(\frac{1}{5}\right)^{5-\mathrm{x}} \cdot \left(\frac{4}{5}\right)^{\mathrm{x}}$

$\mathrm{P}($ **four students are swimmers** $) = \mathrm{P}(\mathrm{X} = 4) = {}^{5}\mathrm{C}_{4} \left(\frac{1}{5}\right) \cdot \left(\frac{4}{5}\right)^{4}$

Q.21 Answer:

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Solution:

A **is independent of itself if and only if**

$$P(A \cap A) = P(A)P(A) \quad \Leftrightarrow \quad P(A) = P(A)^2$$
$$\Leftrightarrow P(A) = 0 \text{ or } P(A) = 1$$

Q.22 Answer:

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Solution:

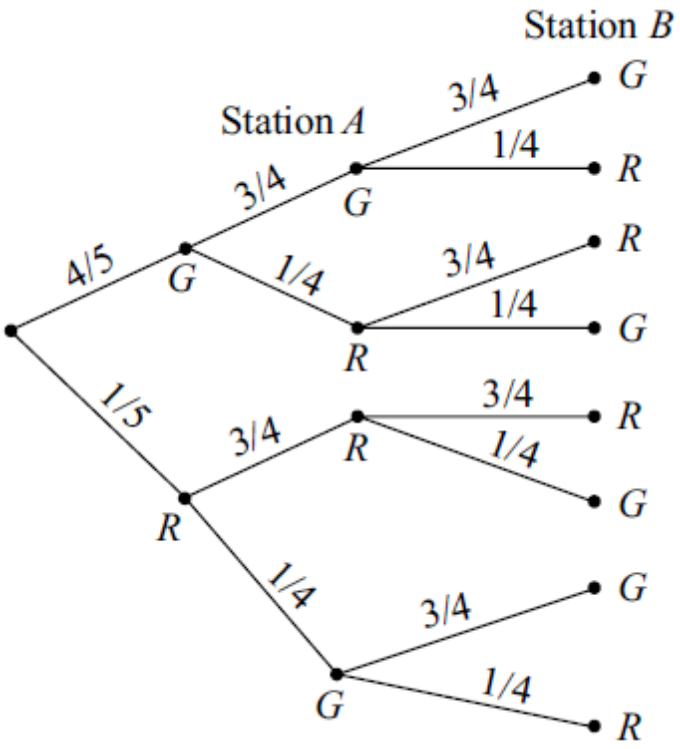
$$P(E) = P(2, 3, 5, 7)$$
$$= 0.23 + 0.12 + 0.20 + 0.07$$
$$= 0.62$$
$$P(F) = P(1, 2, 3) = 0.15 + 0.23 + 0.12$$
$$= 0.5$$
$$P(E \cap F) = P(2, 3) = 0.23 + 0.12 = 0.35$$
$$\text{Thus, } P(E \cup F) = P(E) + P(F) - P(E \cap F)$$
$$= 0.62 + 0.5 - 0.35 = 0.77$$

Q.23 Answer:

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Solution:

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Let G, E_1, E_2, and E denote the following events:

G : Original signal is green E_1 : A receives the signal correctly E_2 : B receives the signal correctly E : B receives the green signal

We have

$$
\begin{aligned}
E =& GE_1E_2 \cup GE_1'E_2' \cup G'E_1E_2' \cup G'E_1'E_2 \\
\Rightarrow P(E) =& P\left(GE_1E_2\right) + P\left(GE_1'E_2'\right) \\
&+ P\left(\mathrm{G}'E_1E_2'\right) + P\left(G'E_1'E_2'\right) \\
&= \left(\frac{4}{5}\right)\left(\frac{3}{4}\right)\left(\frac{3}{4}\right) + \left(\frac{4}{5}\right)\left(\frac{1}{4}\right)\left(\frac{1}{4}\right) \\
&\quad + \left(\frac{1}{5}\right)\left(\frac{3}{4}\right)\left(\frac{1}{4}\right) + \left(\frac{1}{5}\right)\left(\frac{1}{4}\right)\left(\frac{3}{4}\right) \\
&= \frac{36+4+3+3}{80} = \frac{23}{40}
\end{aligned}
$$

$$
\begin{aligned}
\text{Also,} \quad P(G \cap E) &= P\left(GE_1E_2\right) + P\left(GE_1'E_2'\right) \\
&= \frac{40}{80} = \frac{1}{2}
\end{aligned}
$$

$$
\therefore \quad P(G \mid E) = \frac{P(G \cap E)}{P(E)} = \frac{1/2}{23/40} = \frac{20}{23}
$$

Q.24 Answer:

—

Solution:

$$
\begin{aligned}
&P(B) = 1 - P\left(B'\right) = 1 - 6/36 = 5/6 \\
&P(A \mid B) = \tfrac{P(A \cap B)}{P(B)} = \tfrac{P\{(6,5)\}}{P(B)} = \tfrac{1/36}{5/6} = \tfrac{1}{30}
\end{aligned}
$$

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Q.25 Answer:

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Solution:

Total number of cases $n(S) = 6^3 = 216$

Product is prime only when two outcomes are 1 and the third is prime i.e. $2, 3, 5$

If it is $2, 1, 1$**, then the number of cases is** $3.$

Similarly, 3 cases for $3, 1, 1$ **and** $5, 1, 1$ **each Hence, favorable cases** $= 9$

Hence, required probability $p = \frac{1}{24}$

$\Rightarrow \frac{1}{4p} = 6$

Q.26 Answer:

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Solution:

Let there be x **red socks and** y **blue socks. Let** $x > y$

$$\text{Then } \frac{x}{\frac{x}{C_2+y}C_2}x+y'C_2 = \frac{1}{2}$$

$$\Rightarrow \quad \frac{x(x-1)+y(y-1)}{(x+y)(x+y-1)} = \frac{1}{2}$$

$$\Rightarrow \quad 2x^2 - 2x + 2y^2 - 2y = x^2 + 2xy + y^2 - x - y$$

$$\Rightarrow \quad x^2 - 2xy + y^2 = x + y$$

$$\Rightarrow \quad (x-y)^2 = x + y$$

$$\Rightarrow \quad |x-y| = (x+y)^{1/2}$$

Since $x + y \leq 17, x - y \leq \sqrt{17}$

As $x - y$ **must be an integer** $\Rightarrow x - y = 4$

$\therefore \quad x + y = 16$

Solving, we get $x = 7$

Q.27 Answer:

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Solution:

Let E**, denote an event when a coin with two heads is selected and** E_2 **an event when a fair coin is selected.**

Let A **be the event when the toss results in heads. Then,** $P(E_1)$

$= \mathbf{1}/(n+1), P(E_2) = 1/(n+1), P(A/E_1) = 1$ and

$P(\boldsymbol{A}/E_2) = 1/2.$

$$\therefore P(\boldsymbol{A}) = P(E_1)\,P(A/E_1) + P(E_2)\,P(A/E_2)$$

$$\Rightarrow \frac{7}{12} = \frac{1}{n \times 1} \times 1 + \frac{n}{n+1} \times \frac{1}{2}$$

$$\Rightarrow 12 + 6n = 7n + 7$$

$$\Rightarrow n = 5$$

Q.28 Answer:

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Solution:

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We have $n = 16, npq = 2^2 = 4$

$\therefore \quad p(1-p) = 1/4 \quad \Rightarrow \quad 4p^2 - 4p + 1 = 0$

$\Rightarrow \quad (2p-1)^2 = 0 \Rightarrow \quad p = 1/2$

Thus, $\quad \text{mean} = np = 16(1/2) = 8$

Q.29 Answer:

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Solution:

$E(X) = 1\left(\frac{1}{2}\right) + 2\left(\frac{1}{5}\right) + 3\left(\frac{3}{25}\right)$
$+(2k)\left(\frac{1}{10}\right) + (3k)\left(\frac{1}{25}\right) + (5k)\left(\frac{1}{25}\right)$
$= \frac{1}{50}[25 + 20 + 18 + 10k + 6k + 10k]$
$\Rightarrow \quad 2.3 = \frac{1}{50}(63 + 26k) \Rightarrow 115 = 63 + 26k$
$\Rightarrow \quad k = 2$

Q.30 Answer:

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Solution:

Yes, the player ranked number 1 will win all of his matches, and will eventually win the tournament. probability is that the players ranked 1 and 2 will meet in the final (because only in this scenario, the ranked 1 player will win, and the ranked 2 player will be the runner-up),for the two players to meet in the final, we need to avoid that they meet before the final

In round 1,1 can meet all 31 player except $2, \mathrm{p} = 31/3^2$

In round 2, 1 can meet all 14 player except $2, p = 14/15$

In round 3,1 can meet all 6 player except 2, $p = 6/7$

In round 4,1 can meet all 2 player except $2, \mathrm{p} = 2/3$

Therefore required probability $= \left(31/3^2\right)^* (14/15)^*(6/7)^*(2/3) = 16/31$

$2/\mathrm{p} = 2/(16/31) = 3 \cdot 7$

$[2/\mathrm{p}] = 3$

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